Econometrics III
Bachelor in Econometrics and Data Science
Bachelor in Econometrics and Operations Research

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Weeks 2-3
Panel Data Models

VRIJE UNIVERSITEIT AMSTERDAM

This Course: Econometrics 3

1. Multiple equations in regression models (week 1)

2. Panel data models (weeks 2 and 3)
   - ▷ Introduction and Random Effects model
   - ▷ Fixed Effects model and some extensions

3. Vector autoregressive models (weeks 4 and 5)
   - ▷ Stable VAR(1), dynamic properties, forecasting
   - ▷ Granger causality, impulse response, estimation, Stable VAR(p)

4. Non-stationary, cointegration, vector error correction (week 5)

5. Principal components, dynamic factor models (week 6)

Week 2-3 – Panel Data Models

- Panel Data
- Pooled Estimator
- Random effects model
  - Random effects: between estimation
  - Random effects: within estimation
- Fixed effects model
- Random effects or fixed effects?
- Hausman test
- Recent developments
  - Group-means
  - Two-ways fixed effects model
  - Difference-in-differences method

Econometrics III Assignments

First assignment of this course is made available on Canvas:

- Assignment 1 is on panel data models: `Ectr3_assignment_1_2026.pdf`
- It contains a pre-assignment part for which `Python` code will be made available
- It covers material from Weeks 2 and 3
- The deadline for delivery on Canvas is Friday, 27 February 23:59 hours

The assignments are done by groups of four students.
The composition of a group must be confirmed via Canvas.

Econometrics III Assignments

The deadlines for the three assignments of this course are:

- Assignment 1 is on panel data models:
  `Ectr3_assignment_1_2026.pdf`
  deadline Friday, 27 February 23:59 hours
- Assignment 2 is on vector autoregressive models:
  `Ectr3_assignment_2_2026.pdf`
  deadline Friday, 13 March 23:59 hours
- Assignment 3 is on cointegration and vector error correction models: `Ectr3_assignment_3_2026.pdf`
  deadline Friday, 20 March 23:59 hours

Notice: the Exam is planned for Friday 27 March.

Panel data

Panel data consist of cross-sectional measurements which are observed at different time points.

These two dimensions “cross-section” and “time” make it a

panel data set.

We typically have many subjects in the cross-section (measurements from a survey) and a few repeated measurements over time (surveys can be big and expensive operations).

Panel data example

	A	B	C	D	E	F	G	H	I	J
1	week	exp	wks	bluecol	ind	south	smsa	married	gender	union
2	1	3	32	0	0	1	0	1	0	0
3	2	4	43	0	0	1	0	1	0	0
4	3	5	40	0	0	1	0	1	0	0
5	4	6	39	0	0	1	0	1	0	0
6	5	7	42	0	1	1	0	1	0	0
7	6	8	35	0	1	1	0	1	0	0
8	7	9	32	0	1	1	0	1	0	0
9	1	30	34	1	0	0	0	1	0	0
10	2	31	27	1	0	0	0	1	0	0
11	3	32	33	1	1	0	0	1	0	1
12	4	33	30	1	1	0	0	1	0	0
13	5	34	30	1	1	0	0	1	0	0
14	6	35	37	1	1	0	0	1	0	0
15	7	36	30	1	1	0	0	1	0	0
16	1	6	50	1	1	0	0	1	0	1
17	2	7	51	1	1	0	0	1	0	1
18	3	8	50	1	1	0	0	1	0	1

Panel data from Peru

	A	B	C	D	E	F	G	H	I
1	City	Region	Item	Item group	Item type	Calories	1/1/2003	2/1/2003	3/1/2003
2	2	1	banana	Fruit	perishable	highc	0.45	0.49	0.54
3	6	1	banana	Fruit	perishable	highc	0.81	0.78	0.76
4	12	1	banana	Fruit	perishable	highc	1.03	0.97	0.9
5	13	1	banana	Fruit	perishable	highc	0.7	0.7	0.71
6	19	1	banana	Fruit	perishable	highc	1.63	2.04	1.99
7	23	1	banana	Fruit	perishable	highc	0.64	0.62	0.61
8	3	2	banana	Fruit	perishable	highc	0.84	0.71	0.61
9	4	2	banana	Fruit	perishable	highc	1.07	1.07	1.07
10	7	2	banana	Fruit	perishable	highc	1.49	1.49	1.52
11	17	2	banana	Fruit	perishable	highc	0.73	0.7	0.69
12	20	2	banana	Fruit	perishable	highc	1.16	1.16	1.16
13	22	2	banana	Fruit	perishable	highc	3.11	2.91	2.62
14	5	3	banana	Fruit	perishable	highc	0.83	0.82	0.83
15	8	3	banana	Fruit	perishable	highc	0.95	0.96	0.96
16	9	3	banana	Fruit	perishable	highc	1	1	1
17	10	3	banana	Fruit	perishable	highc	0.7	0.66	0.71
18	11	3	banana	Fruit	perishable	highc	1.07	1.03	1.05
19	14	3	banana	Fruit	perishable	highc	0.62	0.66	0.61
20	18	3	banana	Fruit	perishable	highc	0.85	0.71	0.85
21	1	4	banana	Fruit	perishable	highc	0.45	0.48	0.49
22	15	4	banana	Fruit	perishable	highc	0.52	0.57	0.54
23	16	4	banana	Fruit	perishable	highc	1.04	1.04	1.04
24	21	4	banana	Fruit	perishable	highc	0.83	0.85	0.88
25	24	4	banana	Fruit	perishable	highc	0.92	0.88	0.9
26	2	1	carrot	vegetables	perishable	lowc	0.73	0.73	0.58
27	6	1	carrot	vegetables	perishable	lowc	0.98	0.87	0.75
28	12	1	carrot	vegetables	perishable	lowc	0.86	0.8	0.71
29	13	1	carrot	vegetables	perishable	lowc	0.71	0.73	0.82

Panel data model

Notation is not very different as elsewhere: $y_{it} = \widetilde{x}_{it}\beta + \varepsilon_{it}$

We have dependent variable y_{it} for subject i at time period t, where $i = 1, \ldots, N$ and $t = 1, \ldots, T$.

Usually, N is *much* larger than T: $N >> T$.

Further, we have k explanatory variables in vector $\widetilde{x}_{it}$, that is

$$\widetilde{x}_{it} = \left(X_{it}^1, X_{it}^2, \ldots, X_{it}^k\right).$$

Some variables vary both with i and t, other variables vary with i only or with t only. This can lead to some practical issues.

The noise term is denoted by ε_{it}.

Panel data model

We develop the notation further. We have the "pooled" model $y_i = X_i\beta + \varepsilon_i$ with

$$y_i = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{pmatrix}, \; X_i = \begin{pmatrix} X_{i1}^1 & X_{i1}^2 & \cdots & X_{i1}^k \\ X_{i2}^1 & X_{i2}^2 & \cdots & X_{i2}^k \\ \vdots & \vdots & \ddots & \vdots \\ X_{iT}^1 & X_{iT}^2 & \cdots & X_{iT}^k \end{pmatrix}, \; \varepsilon_i = \begin{pmatrix} \varepsilon_{i1} \\ \varepsilon_{i2} \\ \vdots \\ \varepsilon_{iT} \end{pmatrix}.$$

Next we can stack $y_1, \ldots, y_N$ into a model $y = X\beta + \varepsilon$ with

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix}.$$

Panel data regression

We can express the panel data model as a normal regression model

$$y_{it} = \widetilde{x}_{it}\beta + \varepsilon_{it} \qquad \Leftrightarrow \qquad y = X\beta + \varepsilon,$$

where $\widetilde{x}_{it} = (X_{it}^1, X_{it}^2, \ldots, X_{it}^k)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_k)'$.

In the basic case of independent and identically distributed (IID) errors, we can just apply OLS (pooled estimator).

$$\widehat{\beta} = (X'X)^{-1}X'y.$$

Nothing special here, but make sure you have correct dimensions!

However, $\widehat{\beta}$ has ignored the panel structure of the data completely.
We may expect that the observations from the same subject i are more correlated compared to those from another subject, even if observations from subject i are observed at different time points.

Panel noise structure

To consider this panel data feature in the model, we enforce structure on the noise:

$$\varepsilon_{it} = \alpha_i + \eta_{it},$$

where α_i and η_{it} are IID noise terms which have mean zero and variances σ_α^2 and σ_η^2, respectively.

First, we assume that

$$\mathbb{E}(\alpha_i \eta_{it}) = 0, \qquad \mathbb{E}(\widetilde{x}_{it} \eta_{it}) = 0, \qquad \forall i, t.$$

Hence, α_i is the "subject" i effect in the model.

Panel noise structure

In the panel noise decomposition,

$$\varepsilon_{it} = \alpha_i + \eta_{it},$$

we either assume that

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0, \quad \text{(random effects)},$$

or

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) \neq 0, \quad \text{(fixed effects)}.$$

These two different assumptions imply different methods for estimation.

Random effects and fixed effects

Notice: the focus is on the role of the *subject effect* α_i which is part of the model noise $\varepsilon_{it} = \alpha_i + \eta_{it}$.

When we can assume that α_i does not coincide with any variation in X, we can truly treat α_i as a random effect in y: when assuming

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0, \quad \text{we treat } \alpha_i \text{ as a random effect.}$$

When this assumption cannot be made, we need to account for it, and we can treat α_i as a fixed, unknown effect in y: when assuming

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) \neq 0, \quad \text{we treat } \alpha_i \text{ as a fixed effect.}$$

Random effects model

We first consider the case of the random effect model in which we can regard both α_i and η_{it} as proper noise terms.

In this case, standard OLS regression estimate is consistent, BUT it is not efficient (it is not the best!).

We need to account for the panel noise structure and apply GLS to obtain efficient estimates !

Given that the variances σ_α^2 and σ_η^2 are typically unknown, we rely on *Feasible GLS*.

GLS for Random effects model

The "random effect" can be represented as

$$\begin{aligned} y_{it} &= \widetilde{x}_{it}\beta + \varepsilon_{it}, \qquad \varepsilon_{it} = \alpha_i + \eta_{it}, \\ &\text{or as} \\ y_i &= X_i\beta + \varepsilon_i, \qquad \varepsilon_i = \alpha_i 1_T + \eta_i, \end{aligned}$$

where α_i is a scalar, 1_T is a $T \times 1$ vector of ones, and $\eta_i = (\eta_{i1}, \ldots, \eta_{iT})'$, for $i = 1, \ldots, N$.

Next, we establish the (non-diagonal) variance matrix for ε_i, that is

$$\Sigma := \mathbb{V}ar(\varepsilon_i) = \sigma_\eta^2 I_T + \sigma_\alpha^2 1_T 1_T',$$

and, for ε, that is

$$\Omega := \mathbb{V}ar(\varepsilon) = I_N \otimes \Sigma.$$

GLS for Random effects model

The GLS applied to the panel model

$$y_i = X_i\beta + \varepsilon_i, \qquad \varepsilon_i \sim \mathrm{NID}(0, \Sigma),$$

for $i = 1, \ldots, N$, with $T \times T$ variance matrix

$$\Sigma = \sigma_\eta^2 I_T + \sigma_\alpha^2 1_T\, 1_T',$$

is given by

$$\widehat{\beta}_{\mathrm{GLS}} = \left(\sum_{i=1}^{N} X_i' \Sigma^{-1} X_i \right)^{-1} \sum_{i=1}^{N} X_i' \Sigma^{-1} y_i.$$

GLS in practice

For matrix $\Sigma = \sigma_\eta^2 I_T + \sigma_\alpha^2 1_T\, 1'_T$, it can be shown that

$$\Sigma^{-\frac{1}{2}} = \frac{1}{\sigma_\eta}\left[I_T - \frac{(1-\theta)}{T} 1_T 1'_T\right],$$

with

$$\theta^2 = \frac{\sigma_\eta^2}{T\sigma_\alpha^2 + \sigma_\eta^2}.$$

such that

$$\Sigma^{-\frac{1}{2}} \times \Sigma^{-\frac{1}{2}} = \Sigma^{-1}$$

Result can be proven based on a particular *matrix inversion lemma*, sometimes referred to as the *Woodbury matrix identity*.

GLS in practice

By having expressions for Σ and $\Sigma^{-\frac{1}{2}}$, we can carry out the GLS estimation

$$\widehat{\beta}_{\mathsf{GLS}} = \left(\sum_{i=1}^{N} X_i' \Sigma^{-1} X_i \right)^{-1} \sum_{i=1}^{N} X_i' \Sigma^{-1} y_i,$$

via transformations $X_i^* = \Sigma^{-\frac{1}{2}} X_i$ and $y_i^* = \Sigma^{-\frac{1}{2}} y_i$, and applying OLS:

$$\widehat{\beta}_{\mathsf{GLS}} = \left(\sum_{i=1}^{N} X_i^{*\prime} X_i^* \right)^{-1} \sum_{i=1}^{N} X_i^{*\prime} y_i^*.$$

GLS in practice

The GLS requires values for σ_η^2 and σ_α^2:
otherwise we cannot compute Σ nor $\Sigma^{-\frac{1}{2}}$.

There are various ways to jointly estimate β and the two unknown variances σ_η^2 and σ_α^2. For example, we can adopt full *maximum likelihood estimation* (MLE) but this method is somewhat involved, conceptually and numerically.

Most common solution is to consider feasible regression by means of *between* and *within* estimation.

We discuss these two methods next.

Between estimation: average over t

By averaging the elements of the data vector y_i over t, we "average out" the randomness in y_i caused by the noise η.
We define

$$\overline{y}_i = \frac{1}{T}\sum_{t=1}^{T} y_{it}, \qquad \overline{X}_i = \frac{1}{T}\sum_{t=1}^{T} X_{it},$$

and an obvious regression model for $\overline{y}_i$ is given by

$$\overline{y}_i = \overline{X}_i\beta + \overline{\varepsilon}_i, \qquad i = 1, \dots, N,$$

with the noise reducing to

$$\overline{\varepsilon}_i = \frac{1}{T}\sum_{t=1}^{T} \varepsilon_{it} = \alpha_i + \frac{1}{T}\sum_{t=1}^{T} \eta_{it} \approx \alpha_i,$$

for large enough T, based on the *law of large numbers* (LLN).

Apply OLS on averages

We then have the approximating model

$$\overline{y}_i = \overline{X}_i\beta + \alpha_i, \qquad \alpha_i \sim \text{NID}(0, \sigma^2_\alpha), \qquad i = 1, \ldots, N.$$

Estimate of β is obtained by applying the OLS method, and estimate of σ^2_α is obtained from OLS residuals.
There is an insight that applying OLS on $(\overline{y}, \overline{X})$, is effectively applying *two-stage least squares* (2SLS) on the original data (y, X).

Notice: when applying OLS for a standard regression $y = X\beta + \varepsilon$, we define the hat-maker P and the residual-maker M matrices as

$$P = X(X'X)^{-1}X', \qquad M = I - P = I - X(X'X)^{-1}X'.$$

P, M idempotent matrices: $P = P'$, $P^2 = P$, $M = M'$, $M^2 = M$.
Also, $\widehat{y} = Py$, $e = My$, $PX = X$, $MX = 0$, $PM = MP = 0$.

Apply OLS on averages: 2SLS

We then have the approximating model

$$\bar{y}_i = \overline{X}_i \beta + \alpha_i, \qquad \alpha_i \sim \text{NID}(0, \sigma_\alpha^2), \qquad i = 1, \ldots, N.$$

An estimate of β can be obtained by OLS on this model which is *in-effect* equivalent to *two-stage least squares* on the original data:

1. transform the data

   $$\bar{y} = P_D y, \qquad \overline{X} = P_D X,$$

   where $P_D = D(D'D)^{-1}D'$ and $D = I_N \otimes 1_T$;
2. compute the between estimator

   $$\widehat{\beta}_B = (\overline{X}'\overline{X})^{-1}\overline{X}'\bar{y} = (X'P_D X)^{-1}X'P_D y.$$

Matrix P_D : numerical example

Example: we have $N = 4$ and $T = 3$, then matrix D is given by

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{such that} \quad D\,\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \\ \alpha_2 \\ \alpha_3 \\ \alpha_3 \\ \alpha_3 \\ \alpha_4 \\ \alpha_4 \\ \alpha_4 \end{pmatrix},$$

Based on this matrix D we can construct matrix P_D.

Matrix P_D : numerical example

The idempotent "between" matrix is $P_D = D(D'D)^{-1}D'$ where

$$D'D = \begin{bmatrix} T & 0 & 0 & 0 \\ 0 & T & 0 & 0 \\ 0 & 0 & T & 0 \\ 0 & 0 & 0 & T \end{bmatrix}, \quad \text{and} \quad (D'D)^{-1} = T^{-1} I_N,$$

where $N = 4$ and $T = 3$, in our example. It follows that

$$(D'D)^{-1}D'y = \begin{pmatrix} \overline{y}_1 \\ \overline{y}_2 \\ \overline{y}_3 \\ \overline{y}_4 \end{pmatrix},$$

and that $P_D y$ provides duplicates of the means $\overline{y}_i$ for y.

Matrix P_D

We should notice that this matrix P_D is a powerful matrix !
It transforms the $NT \times 1$ vector y into a $NT \times 1$ vector $\overline{y} = P_D y$ with T duplicates of averages (over time t) of all $T \times 1$ vectors y_i, for $i = 1, \ldots, N$.

Hence, P_D is the "average maker". It duplicates the ith average T times for each i, but this does not lead to numerical issues; it is convenient that data vectors y and data matrices X remain of the same dimensions.

The projection matrix P_D is idempotent such that $P_D = P_D'$ and $P_D\, P_D = P_D$. We have used these properties to obtain the expression for $\widehat{\beta}_B$. Please verify !

Between estimation completed

We have the approximating model

$$\overline{y}_i = \overline{X}_i\beta + \alpha_i, \qquad \alpha_i \sim \mathrm{NID}(0, \sigma_\alpha^2), \qquad i = 1, \ldots, N,$$

and the between estimate $\widehat{\beta}_B = (X'P_D X)^{-1} X'P_D y$ from which the between regression residuals (scalars)

$$a_i = \overline{y}_i - \overline{X}_i\widehat{\beta}_B,$$

can be used to estimate σ_α^2, that is

$$\widehat{\sigma}_\alpha^2 = \frac{1}{N-k}\sum_{i=1}^{N} a_i^2.$$

The first unknown variance in Σ is now estimated.

Between estimation: standard errors

To compute t-tests and to test hypotheses related to β, after between estimation, we require the standard error of $\widehat{\beta}_B$. The between estimator is

$$\widehat{\beta}_B = (\overline{X}'\overline{X})^{-1}\overline{X}'\overline{y} = (X'P_DX)^{-1}X'P_Dy.$$

We have

$$\mathrm{Var}(\widehat{\beta}_B) = s_B^2\Sigma_B, \qquad \Sigma_B = (\overline{X}'\overline{X})^{-1} = (X'P_DX)^{-1},$$

where

$$s_B^2 = \widehat{\sigma}_\alpha^2 = \frac{1}{N-k}\sum_{i=1}^{N} a_i^2.$$

The standard error of $\widehat{\beta}_B$ is given by

$$\mathrm{s.e}(\widehat{\beta}_B) = s_B\left[\mathrm{diag}(\Sigma_B)\right]^{1/2},$$

where $\mathrm{diag}(A)$ is a vector with diagonal elements of matrix A.

Within estimation: deviations from averages

To estimate σ_η, the second unknown variance in Σ, we may now eliminate α_i from ε_{it} so that η_{it}*remains*. The suggestion is to consider deviations from the means (or averages).

For this purpose, we consider

$$y_i - 1_T \overline{y}_i = (X_i - \overline{X}_i)\beta + \varepsilon_i - 1_T \overline{\varepsilon}_i,$$

where

$$\varepsilon_i - 1_T \overline{\varepsilon}_i \approx \varepsilon_i - 1_T \alpha_i = \eta_i = (\eta_{i1}, \ldots, \eta_{iT})',$$

for large enough T, and $\overline{y}_i$ is as usually defined as

$$\overline{y}_i = \frac{1}{T} \sum_{t=1}^{T} y_{it}.$$

Apply OLS on deviations: 2SLS

From the resulting approximating regression model

$$\widetilde{y}_i = \widetilde{X}_i\beta + \eta_i,$$

where $\widetilde{y}_i = y_i - 1_T\overline{y}_i$ and $\widetilde{X}_i = X_i - \overline{X}_i$, we can also apply OLS to obtain an estimate of β. Again, it is in effect a 2SLS regression method:

1. transform the data

$$\widetilde{y} = M_D y, \qquad \widetilde{X} = M_D X,$$

   where $M_D = I_{NT} - D(D'D)^{-1}D'$ and $D = I_N \otimes 1_T$;
2. compute the *within estimator*

$$\widehat{\beta}_W = (\widetilde{X}'\widetilde{X})^{-1}\widetilde{X}'\widetilde{y} = (X'M_D X)^{-1}X'M_D y.$$

Matrix M_D : numerical example

The idempotent "within" matrix is $M_D = I_{NT} - D(D'D)^{-1}D'$ where

$$D'D = \begin{bmatrix} T & 0 & 0 & 0 \\ 0 & T & 0 & 0 \\ 0 & 0 & T & 0 \\ 0 & 0 & 0 & T \end{bmatrix}, \quad \text{and} \quad (D'D)^{-1} = T^{-1} I_N,$$

where $N = 4$ and $T = 3$, in our example. It follows that

$$(D'D)^{-1}D'y = \begin{pmatrix} \overline{y}_1 \\ \overline{y}_2 \\ \overline{y}_3 \\ \overline{y}_4 \end{pmatrix},$$

and that $M_D y$ produces deviations of y from the means in $\overline{y}$.

Matrix M_D

Also matrix M_D is a very powerful matrix !
It transforms the $NT \times 1$ vector y into a $NT \times 1$ vector $\widetilde{y} = M_D y$ with data deviations from the average (over time t) of all $T \times 1$ vectors y_i, for $i = 1, \ldots, N$.

Hence, M_D is the "average-deviation maker". It substracts the ith average T times from all elements of y_i, for each i.

The projection matrix M_D is also idempotent such that $M_D = M_D'$ and $M_D\, M_D = M_D$. We have used these properties to obtain the expression for $\widehat{\beta}_W$. Please verify !

Within estimation completed

We have the approximating model

$$\widetilde{y}_i = \widetilde{X}_i \beta + \eta_i, \qquad \eta_i \sim \mathrm{NID}(0, \sigma_\eta^2 I_T), \qquad i = 1, \ldots, N,$$

and the within estimate $\widehat{\beta}_W$ from which the $T \times 1$ vector of within regression residuals

$$u_i = \widetilde{y}_i - \widetilde{X}_i \widehat{\beta}_W,$$

can be used to estimate σ_η^2, that is

$$\widehat{\sigma}_\eta^2 = \frac{1}{N(T-1)-k} \sum_{i=1}^{N} u_i' u_i.$$

We notice that the degrees of freedom is based on $N(T-1)$, rather than NT, as N different means are subtracted from the data: $\mathrm{rank}(M_D) = NT - N = N(T-1)$.

The second unknown variance in Σ is estimated: go Feasible GLS.

Within estimation: standard errors

To compute t-tests and to test hypotheses related to β, after within estimation, we require the standard error of $\widehat{\beta}_W$. The within estimator is

$$\widehat{\beta}_W = (\widetilde{X}'\widetilde{X})^{-1}\widetilde{X}'\widetilde{y} = (X'M_DX)^{-1}X'M_Dy.$$

We have

$$\mathrm{Var}(\widehat{\beta}_W) = s_W^2\Sigma_W, \qquad \Sigma_W = (\widetilde{X}'\widetilde{X})^{-1} = (X'M_DX)^{-1},$$

where

$$s_W^2 = \widehat{\sigma}_\eta^2 = \frac{1}{N(T-1)-k}\sum_{i=1}^{N} u_i'u_i.$$

The standard error of $\widehat{\beta}_W$ is given by

$$\mathrm{s.e}(\widehat{\beta}_W) = s_W\left[\mathrm{diag}(\Sigma_W)\right]^{1/2},$$

where $\mathrm{diag}(A)$ is a vector with diagonal elements of matrix A.

Within estimation with time-invariant Xs

In case the panel model contains explanatory variables that are constant over time, the within estimation method needs some modification.

For example, when jth element of $x_{i,t}$ is $x_{i,t}^{(j)} = c_i$, for some value c_i and for $t = 1, \ldots, T$, then it follows that $\overline{x}_i^{(j)} = c_i$ as well and $x_{i,t}^{(j)} - \overline{x}_i^{(j)} = 0$ for all i and t.

In such cases, the jth element of vector β cannot be identified and cannot be estimated when based on $x_{i,t} - \overline{x}_i$ as for within estimation.

Hence, we need to drop time-invariant Xs from the model, including the intercept, when applying the within estimation method.

We encounter this issue regularly in empirical studies.

Feasible GLS

By placing the estimates for σ_α^2 and σ_η^2 in Σ, to obtain $\widehat{\Sigma}$, we can carry out Feasible GLS estimation for the panel data model as 2SLS (transformation and OLS):

$$\widehat{\beta}_{\mathsf{FGLS}} = \left(\sum_{i=1}^{N} X_i^{*\prime} X_i^* \right)^{-1} \sum_{i=1}^{N} X_i^{*\prime} y_i^*,$$

where

$$X_i^* = \widehat{\Sigma}^{-\frac{1}{2}} X_i = \widehat{\sigma}_\eta^{-1} \left[X_i - (1-\widehat{\theta}) \overline{X}_i \right],$$

and

$$y_i^* = \widehat{\Sigma}^{-\frac{1}{2}} y_i = \widehat{\sigma}_\eta^{-1} \left[y_i - (1-\widehat{\theta}) 1_T \overline{y}_i \right],$$

with

$$\widehat{\theta}^2 = \frac{\widehat{\sigma}_\eta^2}{T\widehat{\sigma}_\alpha^2 + \widehat{\sigma}_\eta^2}.$$

Feasible GLS: standard errors

To compute t-tests and to test hypotheses related to β, after Feasible GLS, we require the standard error of $\widehat{\beta}_{\mathsf{FGLS}}$. The FGLS estimator is

$$\widehat{\beta}_{\mathsf{FGLS}} = \left(\sum_{i=1}^{N} X_i^{*\prime} X_i^* \right)^{-1} \sum_{i=1}^{N} X_i^{*\prime} y_i^* .$$

We have

$$\mathrm{Var}(\widehat{\beta}_{\mathsf{FGLS}}) = \Sigma_{\mathsf{FGLS}}, \qquad \Sigma_{\mathsf{FGLS}} = \left(\sum_{i=1}^{N} X_i^{*\prime} X_i^* \right)^{-1},$$

The standard error of $\widehat{\beta}_{\mathsf{FGLS}}$ is given by $\mathrm{s.e}(\widehat{\beta}_{\mathsf{FGLS}}) = [\mathrm{diag}(\Sigma_{\mathsf{FGLS}})]^{1/2}$.

Variance adjustment, small T

Both the between and within estimations can be considered as accurate when T is large enough.

In cases where T is really small ($T = 2, 3, 4$), the approximations are clearly not great.

This is especially relevant for the "between" estimator as this is a regression based on N obervations, whatever the value of T. To account for small T, we can consider the original decomposition for $\bar{\varepsilon}_i$ in the "between" regression

$$\bar{\varepsilon}_i = \frac{1}{T}\sum_{t=1}^{T}\varepsilon_{it} = \alpha_i + \frac{1}{T}\sum_{t=1}^{T}\eta_{it},$$

and we leave the last term in.

Variance adjustment, small T

The original decomposition for $\bar{\varepsilon}_i$ in the "between" regression $\bar{\varepsilon}_i = \alpha_i + T^{-1}\sum_{t=1}^{T}\eta_{it}$, has a variance of

$$\mathrm{Var}(\bar{\varepsilon}_i) = \sigma_\alpha^2 + \frac{1}{T^2}\sum_{t=1}^{T}\sigma_\eta^2 = \sigma_\alpha^2 + \frac{1}{T}\sigma_\eta^2.$$

Hence we can consider the estimate

$$\widehat{\sigma}_\alpha^2 = \frac{1}{N}\sum_{i=1}^{N} a_i^2 - \frac{1}{T}\widehat{\sigma}_\eta^2,$$

as being more accurate for σ_α^2 where $\widehat{\sigma}_\eta^2$ is provided by the "within" regression residuals (based on NT residuals, more accurate).

We notice that the last term (the small T adjustment) becomes smaller as T increases.

Random effects: when do assumptions fail?

The random effects model is given by

$$y_{it} = \widetilde{x}_{it}\beta + \varepsilon_{it}, \qquad \varepsilon_{it} = \alpha_i + \eta_{it},$$

where $\widetilde{x}_{it} = (X_{it}^1, X_{it}^2, \ldots, X_{it}^k)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_k)'$,
with RE assumption

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0.$$

This assumption is often regarded as somewhat unrealistic.
Also RE strategy can have poor performance.

- Unobserved heterogeneity
- Regressors included because of individual effects
- Small T, averaging over t is unreliable

Unobserved heterogeneity

If subjects i differ in ways you cannot observe — ability, risk preference, firm culture, managerial quality — and those factors ALSO influence the regressors, then RE assumption $\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0$ NOT realistic.

Examples:

- Wage equation: $wage_{it} = \ldots + \beta \cdot edu_i + \ldots$, here "individual ability" (α_i) affects both wages and education (edu)
- Firm productivity: $prod_{it} = \ldots + \beta \cdot msize_i + \ldots$, here "managerial quality" (α_i) affects both production and input choices such as management size ($msize$)
- Health: $health_{it} = \ldots + \beta \cdot jogging_{it} + \ldots$, here "innate health" ($\alpha_i$) affects both health outcomes and lifestyle choices such as going out jogging regularly or not ($jogging$)

Fixed Effects model

So we may not be so happy now with the random effects (RE) model

$$y_{it} = \widetilde{x}_{it}\beta + \varepsilon_{it}, \qquad \varepsilon_{it} = \alpha_i + \eta_{it},$$

where $\widetilde{x}_{it} = (X_{it}^1, X_{it}^2, \ldots, X_{it}^k)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_k)'$, with "random effects" assumption $\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0$.

Hence, a more realistic assumption is perhaps the fixed effect (FE) model:

$$\mathbb{E}(\widetilde{x}_{it}\alpha_i) \neq 0.$$

This assumption is often regarded as somewhat more realistic: it counters the arguments above.

However, least squares assumption $\mathbb{E}(\widetilde{x}_{it}\varepsilon_{it}) \neq 0$ is violated, what to do ? Help !

Least squares dummy variable representation

In case of "fixed effects" model with $\mathbb{E}(\widetilde{x}_{it}\alpha_i) \neq 0$, for $t = 1, \ldots, T$, we can not rely on least squares since this leads to biased estimators for β.

We can isolate the troublemaker α_i and transfer it into the model equation:

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha_i + \eta_{it},$$

in matrix notation as before, we have

$$y = X\beta + D\alpha + \eta.$$

where D is a $NT \times N$ matrix: $D = I_N \otimes 1_T$.

This matrix formulation is also known as the *least squares dummy variable* (LSDV) model.

Eliminate the trouble-maker α_i: option I

The two expressions of the fixed effect model can lead to simple solutions to remove α_i from the analysis.

First,

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha_i + \eta_{it},$$

invites us to consider first differences : $\Delta y_{it} := y_{it} - y_{i,t-1}$, then we have

$$\Delta y_{it} = \Delta\widetilde{x}_{it}'\beta + \Delta\eta_{it}, \qquad t = 2, \ldots, T,$$

where α_i is eliminated from the model, at the cost of losing N observations due to differencing.

However, least squares is now valid for this model.

Eliminate the trouble-maker α_i: option II

Second, we can consider the LSDV

$$y = X\beta + D\alpha + \eta,$$

and treat β and α both as regression coefficient vectors.

In this case and under this construction, we can estimate β simultaneously (with α) indirectly by application of the "within estimation", that is

$$\widehat{\beta}_W = (X' M_D X)^{-1} X' M_D\, y.$$

We notice that

$$M_D\, y = M_D X\, \beta + M_D \eta,$$

since $M_D\, D = 0$. Hence, α is eliminated from the regression.

Fixed Effects elimination / estimation

We have provided some solutions to treat the fixed effects model:

- differencing over consecutive time period : Δy_{it},
- treating it as a LSDV model and consider within estimation.

For the LSDV solution, the within estimator is one of many other candidates of fixed effects β estimators: any linear transformation that eliminates α_i will lead to a valid fixed effects estimator.

Note: fixed effects estimation mostly refers to within estimation.

Discussion: Random effects or Fixed effects?

After much work on estimation of β for RE and FE model settings, we wonder what is best ?

Remark 1: The fixed effects estimates are consistent even when the "random effects" specification is correct.

Remark 2: The random effects estimates are not consistent when the "fixed effects" specification holds.

A tentative conclusion: generally, the fixed effects estimator (say $\widehat{\beta}_W$) is to be preferred to the random effects estimator (say $\widehat{\beta}_{\mathsf{FGLS}}$), unless ... we can be certain that we measure all time-invariant factors as regressors, or possibly correlated with the other regressors, such that α_i is not needed in the model.

Hausman test

To gain some insights into the seriousness of “inconsistency” in the panel regression, Hausman (1978) has proposed a statistical test that exploits this difference between the fixed effects and random effects parameter estimates.

Hausman (1978) proposed a test statistic for the hypothesis that the individual effects α_i are not correlated with the Xs: $\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0$.

Under the null hypothesis, we can assume the validity of the “random effects” model and the optimality of “random effects” estimation, feasible GLS. But also the “within” estimation is valid and leads to consistent estimates.

Under the alternative hypothesis, we have the “fixed effects” model.

Hausman test

In the Hausman test, the null hypothesis states that the random effects FGLS estimator is correct with $\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0$.

The Hausman test is given by

$$H = (\widehat{\beta}_W - \widehat{\beta}_{FGLS})'(s_W^2\Sigma_W - \Sigma_{FGLS})^{-1}(\widehat{\beta}_W - \widehat{\beta}_{FGLS}),$$

where $s_W^2\Sigma_W$ and Σ_{FGLS} are the variance matrices of $\widehat{\beta}_W$ and $\widehat{\beta}_{FGLS}$, respectively.

Under the null, both estimates are consistent and distance $|\widehat{\beta}_W - \widehat{\beta}_{FGLS}|$ is "small", we can assume that $\mathbb{E}(\widetilde{x}_{it}\alpha_i) = 0$.

Under the alternative, $\widehat{\beta}_{FGLS}$ is not consistent and this distance is not "small", we have $\mathbb{E}(\widetilde{x}_{it}\alpha_i) \neq 0$.

Note: Hausman test is effectively an exogeneity test !

Hausman test

The Hausman test statistic as given by

$$H = (\widehat{\beta}_{FGLS} - \widehat{\beta}_W)'(\Sigma_W - \Sigma_{FGLS})^{-1}(\widehat{\beta}_{FGLS} - \widehat{\beta}_W),$$

is (aymptotically) distributed as

$$H \sim \chi^2_k \quad (k \text{ is the number of regressors in } \beta).$$

A "large value" for H will lead to a rejection of the null hypothesis: we reject the "random effects" model and the preference is to use "fixed effects" estimation.

A "small value" for H provides motivation to treat the "random effects" model and the use of FGLS.

Hausman test with time-invariant Xs

In case the panel model contains explanatory variables that are constant over time, the within estimation cannot estimate the corresponding regression coefficients (including the intercept).

This problem does not occur in the FGLS estimation method and hence the issue of time-invariant Xs lead to un-equal dimensions for the vectors β and variance matrices Σ in the Hausman test.

The typical solution in software packages is that the corresponding rows and columns in the Hausman matrix $(\Sigma_W - \Sigma_{FGLS})$ consist of zeros, and its "generalized inverse" is considered.

A (more) practical solution is to drop the corresponding values from $\widehat{\beta}_{FGLS}$ and the corresponding rows and columns from Σ_{FGLS} such that their dimensions align with those of $\widehat{\beta}_W$ and Σ_W, respectively.

The resulting Hausman test is valid and is asymptotically $\chi^2_{k^*}$ distributed with k^* the number of explanatory variables that are NOT time-invariant.

Panel data: there is (much) more to say

- Group means
- Two-Way Fixed Effects
- Dif-in-Dif

Group-means: by example

Example for capturing subject effects, we have $N = 4$ and $T = 3$,

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{such that} \quad D\,\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \\ \alpha_2 \\ \alpha_3 \\ \alpha_3 \\ \alpha_3 \\ \alpha_4 \\ \alpha_4 \\ \alpha_4 \end{pmatrix},$$

Based on this matrix D we can construct matrices P_D and M_D.

Group-means: by example

Example now assumes groups for $i = 1, 2, 3$ and for $i = 4$, then

$$D = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}, \quad \text{such that} \quad D\,\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \\ \alpha_2 \end{pmatrix},$$

Also for this matrix D we can construct matrices P_D and M_D, and proceed as normal with the estimations relying on this "grouped" D.

Statistical test

From having "subject" effects to having "group" effects, we effectively impose linear restrictions on $\alpha_1, \alpha_2, \alpha_3$, that is the constraint

$$\alpha_1 = \alpha_2 = \alpha_3,$$

which we can formally test using F-type tests (or χ^2-type) for the linear hypothesis $H_0 : R\alpha = r$.

The construction of such test statistics has been part of Econometrics 1. However, the relience on matrix D can lead to convenient formulas for computing the test statistics.

When group-means can be used, we lose less data, we obtain more efficient estimates, and we often gain more empirical insights.

Panel data: there is (much) more to say

- Group means
- Two-Way Fixed Effects
- Dif-in-Dif

Two-way Fixed Effects model

The one-way fixed effects model is given by

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha_i + \eta_{it},$$

for $i = 1, \ldots, N$ and $t = 1, \ldots, T$.

In case T is sufficiently large, there can be a need to account for changes of the variables over time. The basic default solution is to introduce fixed time effects in the model, that is

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha_i + \tau_t + \eta_{it},$$

where τ_t is the fixed time effect.

Accounting for “two-ways” fixed effects can be somewhat tricky !

Estimation

In the two-way fixed effects model

$$y_{it} = \tilde{x}_{it}'\beta + \alpha_i + \tau_t + \eta_{it},$$

all α_is and τ_ts cannot be uniquely identified. We usually solve this by having the constraint $\tau_1 = 0$ such that all α_is can already be uniquely determined at $t = 1$.

To apply fixed effect estimation ("within estimation"), we carry out 2SLS where data transformation should both eliminate α_i, $i = 1, \ldots, N$ and τ_t, $t = 2, \ldots, T$.

Estimation

In the two-way fixed effects model

$$y_{it} = \widetilde{x}'_{it}\beta + \alpha_i + \tau_t + \eta_{it},$$

we consider a data transformation that should both eliminate α_i, for $i = 1, \ldots, N$ and τ_t, for $t = 2, \ldots, T$.

Suggestion for this transformation is

$$\widehat{y}_{it} = y_{it} - \overline{y}_{i*} - \overline{y}_{*t} + \overline{y}_{**},$$

where $\overline{y}_{i*}$ is defined earlier as $\overline{y}_i$ and

$$\overline{y}_{*t} = \frac{1}{N}\sum_{i=1}^{N} y_{it}, \qquad \overline{y}_{**} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} y_{it}.$$

Two-way Fixed Effects model

This two-ways transformation is equivalent (from a projection's view) to the P_D transformation, but now based on, with $N = 4$ and $T = 3$,

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}, \quad \text{such that} \quad D \begin{pmatrix} \alpha \\ \tau \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_1 + \tau_2 \\ \alpha_1 + \tau_3 \\ \alpha_2 \\ \alpha_2 + \tau_2 \\ \alpha_2 + \tau_3 \\ \alpha_3 \\ \alpha_3 + \tau_2 \\ \alpha_3 + \tau_3 \\ \alpha_4 \\ \alpha_4 + \tau_2 \\ \alpha_4 + \tau_3 \end{pmatrix},$$

We have implicitly set $\tau_1 = 0$ s.t. $\tau = (\tau_2, \tau_3)'$.

Two-way Fixed Effects model

The transformations M_D, with D given above, can be applied to both y and X in a fixed effects panel model.

Then, also for this matrix D, we can proceed as normal with the estimations relying on the transformed data, as in the 2SLS procedure for the within estimation.

Again, we have shown that extensions such as the two-way fixed effects model can be incorporated in a relatively straightforward manner.

Two-way Fixed Effects: Pooled Estimation

In the two-way fixed effects model

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha_i + \tau_t + \eta_{it},$$

we can consider pooling the fixed effects, we then obtain

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha + \tau \cdot t + \eta_{it},$$

where the subject effect is the same for all i, and the time effect is reduced to a time trend with slope coefficient τ.

This "pooled" specification is highly parsimonious, we only have two coefficients α and τ. The pooling can also only take place for one or the other.

Two-way Fixed Effects model

This two-ways transformation is equivalent (from a projection's view) to the P_D transformation, but now based on, with $N = 4$ and $T = 3$,

$$D = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \text{such that} \quad D \begin{pmatrix} \alpha \\ \tau \end{pmatrix} = \begin{pmatrix} \alpha + \tau \\ \alpha + 2\tau \\ \alpha + 3\tau \\ \alpha + \tau \\ \alpha + 2\tau \\ \alpha + 3\tau \\ \alpha + \tau \\ \alpha + 2\tau \\ \alpha + 3\tau \\ \alpha + \tau \\ \alpha + 2\tau \\ \alpha + 3\tau \end{pmatrix},$$

For this matrix D we can construct matrices P_D and M_D.

Panel data: there is (much) more to say

- Group means
- Two-Way Fixed Effects
- Dif-in-Dif

Difference in differences, Dif-in-Dif or DiD

The difference-in-differences method has been used as early as the 1850's by John Snow, then it was called the "controlled before-and-after study".

DiD is typically used to estimate the effect of a specific intervention or treatment (law change, new policy, program implementation) by comparing changes in outcomes over time between a population that is enrolled in a program (intervention or treatment group) and a population that is not (control group).

With some necessary assumptions, DiD can estimate the Average Treatment Effect (ATE) or the *causal effect*.

Dif-in-Dif: Example

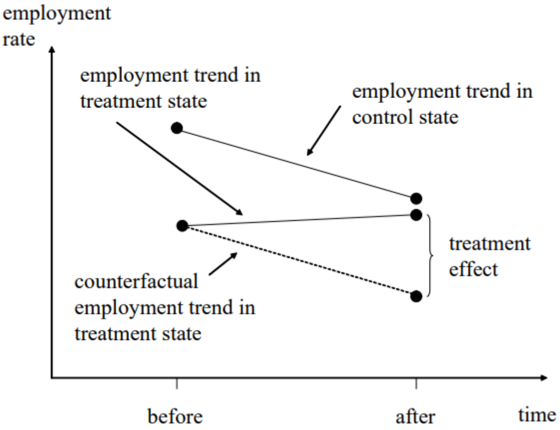
In a famous study of Card and Krueger (1994), they investigated employment rates at fast-food restaurants in two states in the US (PA and NJ).

They studied fast-food employment rates in February 1992 and in November 1992.

Then, in April 1992 a dramatic increase in the NJ state minimum wage took place, the increase did not take place in PA.

The effect of mimimum wage on the employment rate is heavily debated amongst economic policy makers and politicians alike.

Graphical display of treatment effect



(treatment state NJ, control state PA)

Tabular display of treatment effect

$Y_{s,t}$	$s = 0$	$s = 1$	difference
$t = 1$	y_{01}	y_{11}	$y_{01} - y_{11}$
$t = 2$	y_{02}	y_{12}	$y_{02} - y_{12}$
change	$y_{01} - y_{02}$	$y_{11} - y_{12}$	$(y_{01} - y_{11}) - (y_{02} - y_{12})$

Last row, last column entry is treatment or DiD effect.

Other examples

There are famous twin-data sets: differences in salaries can be explained by difference in education (or years of education), given that other relevant (social, family) factors are the same for both, see Ashenfelter and Krueger (1994) and others.

The twin-data sets are also used to study other “big questions”.

DiD is also used for “program evaluations”: does behaviour change after someone/company has gone through a “program”?

Finally, Imbens et al (2001) studied a survey of people playing the lottery in Massachusetts in the mid-1980s, to analyze the effects of the magnitude of lottery prizes on economic behavior.

Treatment effect in panel regression

In a (pooled) two-way fixed effects model

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha + \tau \cdot t + \eta_{it},$$

we can incorporate the treatment effect by introducing the intervention variable S_t where

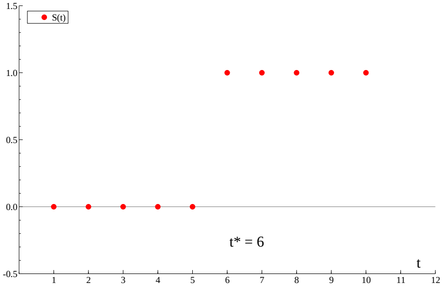
$$S_t = I(t \geq t^*),$$

with indicator function $I(\text{FALSE}) = 0$ and $I(\text{TRUE}) = 1$, for some fixed $t^* \in \{2, \ldots, T\}$, and t^* is the start of the treatment period. The panel regression model

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha + \delta_\alpha S_t + \tau \cdot t + \delta_\tau (S_t \cdot t) + \eta_{it},$$

where δ_τ is the treatment effect coefficient.

Plot of S_t



We have

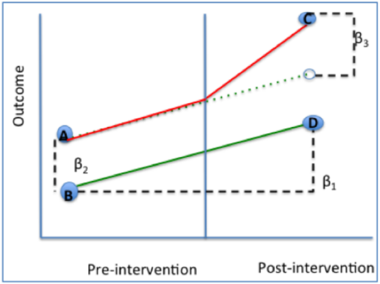
$$S_t = I(t \geq t^*),$$

with indicator function $I(\text{FALSE}) = 0$ and $I(\text{TRUE}) = 1$.

Relation graph and panel regression

$$Y = \beta_0 + \beta_1 t + \beta_2 S_t + \beta_3 (S_t \cdot t) + \ldots + \epsilon$$

Coefficient	Calculation	Interpretation
β_0	B	Baseline average
β_1	D-B	Time trend in control group
β_2	A-B	Difference between two groups pre-intervention
β_3	(C-A)-(D-B)	Difference in changes over time



Treatment effect in panel regression

The DiD panel regression model

$$y_{it} = \widetilde{x}_{it}'\beta + \alpha + \delta_\alpha S_t + \tau \cdot t + \delta_\tau (S_t \cdot t) + \eta_{it},$$

with δ_τ as the treatment effect coefficient.

Given that the focus is much on δ_τ, and possibly also on δ_α, we have

$$y_{it} = \left(\widetilde{x}_{it}',\, S_t\,,\, S_t \cdot t\right)\, (\beta', \delta_\alpha, \delta_\tau)' + \alpha + \tau \cdot t + \eta_{it},$$

where matrix D is designed for the treatment of α and τ, and "within estimation" concentrates on β, δ_α and δ_τ.

Final comment: causal analysis

The DiD method is regarded as a basic tool for causal analysis. It requires a careful setting, various assumptions must be fulfilled to declare *causality* empirically, in contrast to an experimental setting.

Causal analysis is clearly of key importance to policy-makers and it also plays a key role in data science.

Many important developments have been reported. Much research on causality is currently ongoing. The importance of this work is highlighted by Nobel prize winners in this field (Carter, Imbens).

This research should interest you as a young econometrician !!

What did we do?

- We introduced panel data
- We discussed panel error structure
- First we considered the random effects model and FGLS
- Then we question the strong assumptions of the random effects model and consider the fixed effect model and concluded that within estimation is the default method
- Hausman test can be used to verify the random effects assumptions
- Finally we discussed some extensions, including group-means, two-way fixed effects model and difference-in-differences method